Supplementary File of "A Knowledge Extraction-based Evolutionary Algorithm for Large-scale Sparse Multiobjective Optimization Problems"

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I. Mathematical Definitions of Five Sparse Real-world Problems

In this section, five sparse real-world problems are used to verify the performance of our proposed KM-MOEA, including signal reconstruction [1], knapsack problems [2], neural network training problems [3], feature selection problems [4], and vehicle routing problems [5]. Specifically, the mathematical definitions of these five sparse real-world problems are provided as follows.

A. Signal Reconstruction

The signal reconstruction problem [1] aims to minimize the reconstruction error and the sparsity of the signal, as follows:

$$\min \begin{cases} f_1(x) = \frac{|x|}{D} \\ f_2(x) = \|Ax - y\|_2^2 \end{cases} \tag{1}$$

where *x* is the reconstructed signal, *D* is the length of the reconstructed signal, *A* is the sensing matrix, and *y* is the observed signal. f_1 denotes the sparsity of reconstructed signal and f_2 represents the loss of the reconstructed signal.

B. Knapsack Problem

The knapsack problem [2] aims to minimize the total weight and maximize the total value, which is defined as follows:

$$\begin{cases} \max f_1(x) = \sum_{i=1}^{n} v_i x_i \\ \min f_2(x) = \sum_{i=1}^{n} w_i x_i \end{cases} \tag{2}$$

where *x* is the set of all items, $x_i = 1$ denotes the selection of the *i*-th item and $x_i = 0$ indicates the *i*-th item is not selected, v_i stands for the value of the *i*-th item, and w_i is the weight of the *i*-th item. f_1 is the total value of all selected items and f_2 is the total weight of all selected items.

C. Neural Network Training Problem

The neural network training problem [3] aims to minimize the error and complexity of the model, expressed as follows:

$$\min \begin{cases} f_1(x) = \frac{|x|_0}{D} \\ f_2(x) = ErrorRate(x) \end{cases} \tag{3}$$

where *x* is the weights of the neural network, $|x|_0$ is the number of the nonzero weights, *D* is the number of total weights. f_1 is the complexity of the neural network and f_2 is the error rate on training set.

D. Feature Selection Problem

The goal of feature selection problem [4] is to minimize the number of selected features and error rate, as follows:

$$\min \begin{cases} f_1(x) = \sum_{i=1}^{D} x_i \\ f_2(x) = ErrorRate(x) \end{cases} \tag{4}$$

where *x* is the selected features, *D* is the number of all features. f_1 is the number of selected features and f_2 is the error rate.

E. Vehicle Routing Problem

The goal of the multi-objective vehicle routing problem (VRP) [5] is to minimize both the total traveling distance and the number of used vehicles. It can be formulated as follows: The multi-objective CVRP can be formulated using a directed adjacency representation $x_{ij} \in \{0, 1\}$ as

$$\min \begin{cases} f_1(x) = \sum_{i=0}^{N}\sum_{j=0}^{N} d_{ij}\, x_{ij}, \\ f_2(x) = \big|\mathcal{R}(x)\big| \end{cases} \tag{5}$$

subject to

$$\sum_{j=0}^{N} x_{ij} = 1, \qquad \forall\, i = 1, \ldots, N, \tag{6}$$

$$\sum_{i=0}^{N} x_{ij} = 1, \qquad \forall\, j = 1, \ldots, N, \tag{7}$$

$$x_{ii} = 0, \qquad \forall\, i = 0, \ldots, N, \tag{8}$$

$$x_{ij} \in \{0, 1\}, \qquad \forall\, i, j = 0, \ldots, N, \tag{9}$$

$$\sum_{i \in r} q_i \le Q, \qquad \forall\, r \in \mathcal{R}(x), \tag{10}$$

where N is the number of customers. d_{ij} denotes the distance between nodes i and j. $x_{ij} = 1$ if a vehicle travels directly from node i to node j, and 0 otherwise. q_j is the demand of customer j, and Q is the capacity of each vehicle. $f_1(x)$ represents the total traveling distance of all vehicles, while $f_2(x) = |\mathcal{R}(x)|$ represents the number of vehicles used, where $|\mathcal{R}(x)|$ denotes the set of routes (each starting and ending at the depot).

Table A.1: Parameter settings of five application problems

Problems	D	Dataset	Sparsity	Constraints	Objectives	M
SR1	1000	Synthetic [6]	0.19	None	minimize the reconstruction error and the sparsity of the signal	2
SR2	2000	Synthetic [6]	0.18			
SR3	4000	Synthetic [6]	0.15			
KP1	1000	Knapsack [7]	0.19	Capacity constraint	minimize the total weight and maximize the total value	
KP2	2000	Knapsack [7]	0.15			
KP3	4000	Knapsack [7]	0.15			
NN1	435	Connectionist_bench_Sonar [8]	0.22	None	minimize the error and complexity of the model	
NN2	505	MUSK1 [8]	0.18			
NN3	937	$Statlog\_German$ [8]	0.07			
FS1	256	Semeion_handwritten_digit [8]	0.32	None	minimize the number of selected features and the error	
FS2	310	LSVT_voice_rehabilitation [8]	0.01			
FS3	2308	SRBCT [1]	0.03			
VR1	10201	XML100_1111_01 [2]	0.46	Routing and capacity constraints	minimize the traveled distance and the number of vehicles	
VR2	10201	XML100_1111_02 [2]	0.46			
VR3	10201	XML100_1111_03 [2]	0.46			

[1] https://www.kaggle.com/datasets/ahmadalmahsiri/microarray-datasets
[2] https://vrp.atd-lab.inf.puc-rio.br/index.php/en/

Table A.2: IGD results obtained by all compared algorithms on SMOP1-SMOP8 with 2 objectives and 1000-10000 decision variables

Problem	M	D	DKCA [35]	MOEA/PSL [30]	SparseEA [33]	S-ECSO [31]	S-NSGA-II [34]	KE-MOEA
SMOP1	2	1000	1.3596e-2 (2.04e-3) -	1.9783e-2 (5.35e-3) -	3.3319e-2 (4.20e-3) -	7.8711e-1 (0.00e+0) -	1.7234e-2 (3.80e-3) -	**3.7703e-3 (3.06e-5)**
SMOP1	2	2000	2.0661e-2 (2.56e-3) -	1.8803e-2 (3.32e-3) -	4.0444e-2 (9.68e-5) -	4.3576e-1 (0.00e+0) -	2.1016e-2 (6.20e-3) -	**5.7035e-3 (6.38e-4)**
SMOP1	2	4000	2.7148e-2 (9.59e-4) -	2.1475e-2 (4.49e-3) -	4.6704e-2 (2.02e-3) -	7.8582e-1 (0.00e+0) -	2.0120e-2 (2.47e-3) -	**1.0760e-2 (1.87e-3)**
SMOP1	2	10000	3.5271e-2 (1.58e-3) -	2.3697e-2 (7.48e-3) -	4.8806e-2 (9.35e-4) -	6.7250e-2 (0.00e+0) -	2.1779e-2 (2.55e-3) -	**4.7790e-3 (1.03e-4)**
SMOP2	2	1000	3.9727e-2 (6.51e-3) -	6.0770e-2 (5.59e-3) -	9.0841e-2 (2.94e-3) -	8.8292e-1 (0.00e+0) -	4.4975e-2 (1.54e-2) -	**1.3499e-2 (5.21e-3)**
SMOP2	2	2000	5.3666e-2 (2.53e-3) -	6.8523e-2 (1.98e-2) -	9.9901e-2 (8.88e-3) -	5.3602e-1 (0.00e+0) -	7.8536e-2 (2.10e-2) -	**2.1805e-2 (4.43e-3)**
SMOP2	2	4000	7.2069e-2 (2.17e-3) -	6.6513e-2 (4.23e-3) -	1.1681e-1 (2.28e-3) -	8.7983e-1 (0.00e+0) -	8.5890e-2 (9.25e-3) -	**3.3750e-2 (3.64e-3)**
SMOP2	2	10000	8.8486e-2 (2.26e-3) -	6.4705e-2 (7.90e-3) -	1.2300e-1 (2.01e-3) -	1.4107e-1 (0.00e+0) -	1.1563e-1 (6.21e-3) -	**1.9443e-2 (7.42e-4)**
SMOP3	2	1000	1.2316e-2 (1.45e-3) -	2.0757e-2 (4.14e-3) -	4.7572e-2 (3.06e-3) -	7.8711e-1 (0.00e+0) -	2.0200e-2 (5.00e-3) -	**3.7925e-3 (2.99e-5)**
SMOP3	2	2000	1.9156e-2 (1.65e-3) -	2.1095e-2 (2.64e-6) -	5.2464e-2 (1.20e-3) -	4.3576e-1 (0.00e+0) -	2.3176e-2 (1.49e-3) -	**8.1311e-3 (1.14e-3)**
SMOP3	2	4000	2.4187e-2 (1.68e-3) =	4.0825e-1 (8.54e-1) -	5.4797e-2 (1.06e-3) -	7.8582e-1 (0.00e+0) -	2.6852e-2 (5.45e-3) =	**2.4102e-2 (7.21e-3)**
SMOP3	2	10000	3.6587e-2 (5.58e-4) -	3.9706e-2 (1.94e-2) -	5.8483e-2 (7.95e-4) -	6.8018e-2 (0.00e+0) -	2.9293e-2 (4.05e-3) -	**8.6200e-3 (9.14e-4)**
SMOP4	2	1000	4.7944e-3 (1.62e-4) -	5.3489e-3 (4.87e-4) -	4.9747e-3 (2.29e-4) -	4.0370e-3 (0.00e+0) =	4.8469e-3 (1.98e-4) -	**4.1356e-3 (7.93e-5)**
SMOP4	2	2000	4.7116e-3 (3.44e-4) -	4.8517e-3 (2.48e-4) -	4.6888e-3 (6.99e-5) -	4.0692e-3 (0.00e+0) =	4.8488e-3 (2.97e-4) -	**4.0473e-3 (7.45e-5)**
SMOP4	2	4000	4.4134e-3 (1.33e-4) -	5.0346e-3 (2.32e-4) -	4.8462e-3 (1.97e-4) -	4.2029e-3 (0.00e+0) -	4.7315e-3 (1.48e-4) -	**4.1147e-3 (4.80e-5)**
SMOP4	2	10000	4.6958e-3 (1.71e-4) +	**4.7280e-3 (6.03e-5) +**	5.0234e-3 (5.98e-5) +	8.6511e-3 (0.00e+0) +	4.9928e-3 (1.73e-4) +	4.5031e-1 (1.12e-3)
SMOP5	2	1000	7.8795e-3 (2.03e-4) +	8.9783e-3 (2.56e-4) +	7.9077e-3 (1.94e-4) +	8.1932e-1 (0.00e+0) -	**5.4477e-3 (3.20e-4) +**	1.0471e-2 (6.25e-4)
SMOP5	2	2000	7.9462e-3 (5.32e-4) +	8.8830e-3 (8.84e-5) +	**7.7501e-3 (2.59e-4) +**	5.8490e-1 (0.00e+0) -	1.2368e-2 (2.87e-3) =	1.5595e-2 (1.50e-3)
SMOP5	2	4000	8.5922e-3 (2.30e-4) +	8.5108e-3 (2.12e-4) +	**8.1663e-3 (2.74e-4) +**	8.1388e-1 (1.24e-16) -	2.0209e-2 (3.07e-3) +	3.3847e-2 (3.22e-3)
SMOP5	2	10000	9.7245e-3 (2.48e-4) +	**7.7423e-3 (3.43e-4) +**	7.7947e-3 (1.67e-5) +	2.8724e-2 (0.00e+0) +	2.5921e-2 (3.58e-3) +	8.1213e-2 (6.85e-4)
SMOP6	2	1000	9.5029e-3 (1.43e-4) -	9.9227e-3 (8.14e-4) -	1.0585e-2 (1.52e-3) -	8.5338e-1 (0.00e+0) -	5.2552e-3 (3.29e-4) =	**5.1900e-3 (2.64e-4)**
SMOP6	2	2000	9.4012e-3 (4.89e-4) -	1.1195e-2 (1.97e-4) -	1.0216e-2 (3.80e-4) -	6.2139e-1 (0.00e+0) -	**5.7736e-3 (6.35e-4) +**	7.0231e-3 (2.53e-4)
SMOP6	2	4000	1.1158e-2 (3.85e-4) =	1.0217e-2 (3.21e-4) =	1.0871e-2 (2.13e-4) =	8.4942e-1 (1.24e-16) -	**8.7000e-3 (2.32e-3) =**	1.1141e-2 (2.89e-3)
SMOP6	2	10000	1.1559e-2 (2.73e-4) +	8.1316e-3 (7.33e-4) +	**7.9546e-3 (5.85e-5) +**	1.9403e-2 (0.00e+0) +	2.0389e-2 (2.22e-3) =	2.8535e-2 (6.64e-4)
SMOP7	2	1000	5.7891e-2 (3.61e-3) -	7.3846e-2 (1.47e-2) -	1.0326e-1 (1.07e-2) -	8.7173e-1 (0.00e+0) -	7.7384e-2 (2.76e-2) -	**1.7534e-2 (6.54e-3)**
SMOP7	2	2000	7.4546e-2 (2.93e-3) -	5.7318e-2 (9.92e-3) -	1.2093e-1 (8.15e-3) -	4.5995e-1 (0.00e+0) -	1.0483e-1 (2.22e-2) -	**2.4998e-2 (9.26e-3)**
SMOP7	2	4000	9.4917e-2 (5.15e-3) -	9.9471e-2 (8.78e-2) -	1.3732e-1 (3.83e-3) -	8.6916e-1 (1.24e-16) -	1.1803e-1 (2.51e-2) -	**3.9887e-2 (1.84e-2)**
SMOP7	2	10000	1.2097e-1 (2.50e-3) -	6.6418e-2 (1.22e-2) -	1.5487e-1 (1.83e-3) -	2.0124e-1 (0.00e+0) -	1.5380e-1 (4.06e-2) -	**2.6436e-2 (8.68e-4)**
SMOP8	2	1000	1.9085e-1 (1.79e-2) -	3.3384e-1 (5.70e-2) -	2.6721e-1 (1.07e-2) -	1.0774e+0 (0.00e+0) -	1.8237e-1 (2.14e-2) -	**1.1062e-1 (8.10e-3)**
SMOP8	2	2000	2.2075e-1 (1.86e-2) -	2.8362e-1 (7.94e-2)-	3.1644e-1 (1.81e-2) -	1.0769e+0 (0.00e+0) -	1.9802e-1 (1.53e-2)	**1.2590e-1 (6.10e-3)**
SMOP8	2	4000	2.5400e-1 (9.25e-3) -	3.2372e-1 (9.88e-2) -	3.5019e-1 (2.66e-3) -	6.0920e-1 (0.00e+0) -	2.2613e-1 (8.96e-3) -	**1.4989e-1 (6.83e-3)**
SMOP8	2	10000	2.9430e-1 (5.77e-3) -	3.6551e-1 (4.88e-2) -	3.8602e-1 (1.60e-3) -	4.2985e-1 (0.00e+0) -	2.5618e-1 (1.83e-3) -	**1.5051e-1 (4.39e-3)**
+/-/=			6/24/2	6/25/1	6/25/1	3/27/2	5/22/5	

Table A.3: HV results obtained by all compared algorithms on SMOP1-SMOP8 with 2 objectives and 1000-10000 decision variables

Problem	M	D	DKCA [42]	MOEA/PSL [34]	SparseEA [33]	S-ECSO [36]	S-NSGA-II [37]	KE-MOEA
SMOP1	2	1000	5.6677e-01(2.34e-03)-	5.6233e-01(9.03e-03)-	5.0575e-01(2.60e-02)-	4.8563e-01(8.92e-03)-	5.6901e-01(5.17e-03)-	**5.8186e-01(8.07e-05)**
SMOP1	2	2000	5.5773e-01(2.89e-03)-	5.4223e-01(1.60e-02)-	5.0304e-01(2.39e-02)-	4.9045e-01(7.08e-03)-	5.6096e-01(3.00e-03)-	**5.7779e-01(1.04e-03)**
SMOP1	2	4000	5.4962e-01(9.20e-04)-	5.5255e-01(3.87e-03)-	5.2299e-01(3.74e-03)-	4.9873e-01(1.18e-16)-	5.5515e-01(9.68e-03)-	**5.6947e-01(2.16e-03)**
SMOP1	2	10000	5.3936e-01(1.77e-03)-	5.5388e-01(8.16e-03)-	5.2237e-01(1.02e-03)-	4.9904e-01(0.00e+00)-	5.5608e-01(2.84e-03)-	**5.7934e-01(2.07e-04)**
SMOP2	2	1000	5.3333e-01(7.13e-03)-	4.7330e-01(2.53e-02)-	4.1358e-01(3.79e-02)-	3.9540e-01(9.79e-03)-	5.1775e-01(1.12e-02)-	**5.6731e-01(6.63e-03)**
SMOP2	2	2000	5.1584e-01(2.79e-03)-	4.4539e-01(6.82e-02)-	4.0949e-01(3.16e-02)-	3.8460e-01(1.38e-02)-	4.9898e-01(9.27e-03)-	**5.5686e-01(5.48e-03)**
SMOP2	2	4000	4.9216e-01(2.57e-03)-	4.9465e-01(1.40e-02)-	4.3891e-01(1.80e-03)-	4.0281e-01(0.00e+00)-	4.6237e-01(1.15e-03)-	**5.3899e-01(2.64e-03)**
SMOP2	2	10000	4.7131e-01(2.55e-03)-	5.0186e-01(8.86e-03)-	4.2850e-01(2.19e-03)-	4.0714e-01(5.89e-17)-	4.3757e-01(6.51e-03)-	**5.5981e-01(9.26e-04)**
SMOP3	2	1000	5.6850e-01(1.64e-03)-	5.3815e-01(1.54e-02)-	5.0089e-01(1.96e-02)-	4.5452e-01(3.53e-02)-	5.5817e-01(4.63e-03)-	**5.8178e-01(1.32e-04)**
SMOP3	2	2000	5.5977e-01(1.83e-03)-	5.3748e-01(1.57e-02)-	4.9700e-01(1.58e-02)-	4.6873e-01(2.08e-02)-	5.5522e-01(2.50e-03)-	**5.7424e-01(1.58e-03)**
SMOP3	2	4000	5.5315e-01(1.76e-03)=	3.6774e-01(2.76e-01)=	5.1354e-01(5.78e-04)-	0.0000e+00(0.00e+00)-	5.4809e-01(7.67e-03)=	**5.4884e-01(5.61e-03)**
SMOP3	2	10000	5.3753e-01(5.97e-04)-	5.3346e-01(2.14e-02)-	5.0951e-01(8.46e-04)-	4.9794e-01(5.89e-17)-	5.4678e-01(4.54e-03)-	**5.7357e-01(1.22e-03)**
SMOP4	2	1000	8.1842e-01(1.45e-04)-	8.1841e-01(2.53e-04)-	8.1875e-01(1.01e-04)-	7.8261e-01(1.45e-02)-	8.1873e-01(9.02e-05)-	**8.1921e-01(4.97e-05)**
SMOP4	2	2000	8.1853e-01(2.80e-04)-	8.1774e-01(6.83e-04)-	8.1865e-01(1.71e-04)-	8.0697e-01(1.70e-03)-	8.1872e-01(1.01e-04)-	**8.1920e-01(4.37e-05)**
SMOP4	2	4000	8.1878e-01(4.88e-05)-	8.1841e-01(7.94e-05)-	8.1849e-01(9.80e-05)-	8.1441e-01(1.18e-16)-	8.1871e-01(4.94e-05)-	**8.1917e-01(2.34e-05)**
SMOP4	2	10000	8.1861e-01(9.38e-05)+	**8.1866e-01(6.26e-05)+**	8.1838e-01(1.17e-04)+	8.1418e-01(1.18e-16)+	8.1846e-01(1.19e-04)+	3.1040e-01(2.42e-03)
SMOP5	2	1000	**8.1289e-01(2.12e-04)+**	8.1040e-01(1.23e-03)=	7.9529e-01(1.29e-02)=	7.4712e-01(4.47e-03)-	8.1092e-01(9.15e-05)=	8.0959e-01(7.34e-04)
SMOP5	2	2000	**8.1262e-01(6.46e-04)+**	8.1101e-01(1.16e-03)+	7.9206e-01(1.55e-02)=	7.5888e-01(2.06e-02)-	8.1195e-01(3.26e-03)+	8.0380e-01(1.67e-03)
SMOP5	2	4000	8.1183e-01(2.21e-04)+	8.1207e-01(4.39e-04)+	**8.1271e-01(4.38e-04)+**	7.7096e-01(1.18e-16)-	7.9795e-01(3.49e-03)+	7.8152e-01(2.43e-03)
SMOP5	2	10000	8.1041e-01(2.68e-04)+	**8.1298e-01(3.19e-04)+**	8.1285e-01(5.62e-05)+	7.8440e-01(0.00e+00)+	7.9214e-01(3.40e-03)+	7.3049e-01(8.01e-04)
SMOP6	2	1000	8.1075e-01(1.68e-04)-	8.0737e-01(3.15e-03)-	7.8838e-01(1.55e-02)-	7.5556e-01(8.96e-03)-	**8.1716e-01(2.61e-04)+**	8.1644e-01(4.72e-04)
SMOP6	2	2000	8.1079e-01(4.95e-04)-	8.0669e-01(2.19e-03)-	7.8966e-01(1.48e-02)-	7.5383e-01(1.92e-02)-	**8.1594e-01(5.24e-04)+**	8.1379e-01(3.15e-04)
SMOP6	2	4000	8.0873e-01(4.27e-04)+	8.1002e-01(3.95e-04)+	8.0938e-01(5.41e-04)+	7.7597e-01(0.00e+00)-	**8.1341e-01(3.88e-03)+**	8.0648e-01(3.22e-04)
SMOP6	2	10000	8.0824e-01(2.63e-04)+	8.1253e-01(8.00e-04)+	**8.1269e-01(7.24e-05)+**	7.9460e-01(0.00e+00)+	7.9822e-01(2.10e-03)+	7.8959e-01(7.43e-04)
SMOP7	2	1000	2.6976e-01(4.50e-03)-	1.5958e-01(1.19e-01)-	1.3832e-01(5.38e-02)-	1.0136e-01(1.18e-02)-	2.1236e-01(1.06e-02)-	**3.2561e-01(8.85e-03)**
SMOP7	2	2000	2.4629e-01(3.64e-03)-	2.2357e-01(5.64e-02)-	1.3412e-01(4.42e-02)-	1.0062e-01(7.12e-03)-	2.2525e-01(5.89e-03)-	**3.1526e-01(1.27e-02)**
SMOP7	2	4000	2.1732e-01(6.32e-03)-	2.7567e-01(9.96e-04)=	1.6861e-01(9.36e-03)-	1.1358e-01(0.00e+00)-	1.6144e-01(1.22e-02)-	**3.0300e-01(3.16e-02)**
SMOP7	2	10000	1.8486e-01(2.31e-03)-	2.5776e-01(1.49e-02)-	1.5178e-01(1.45e-03)-	1.1354e-01(1.47e-17)-	1.5513e-01(3.03e-02)-	**3.1319e-01(1.15e-03)**
SMOP8	2	1000	1.2131e-01(1.17e-02)-	3.2474e-02(1.83e-02)-	2.4690e-02(2.89e-02)-	3.0269e-03(2.48e-03)-	1.4838e-01(6.49e-03)-	**1.9763e-01(9.73e-03)**
SMOP8	2	2000	9.9914e-02(1.09e-02)-	4.2802e-02(3.93e-02)-	2.3227e-02(2.62e-02)-	4.2428e-03(3.83e-03)-	1.0715e-01(8.83e-03)-	**1.8042e-01(6.46e-03)**
SMOP8	2	4000	7.9136e-02(4.78e-03)-	6.1772e-02(1.09e-02)-	3.4928e-02(2.49e-03)-	1.2527e-02(0.00e+00)-	9.2790e-02(7.68e-03)-	**1.5431e-01(4.41e-03)**
SMOP8	2	10000	5.8189e-02(2.51e-03)-	3.0854e-02(1.30e-02)-	2.2729e-02(3.87e-04)-	1.0714e-02(1.84e-18)-	7.8417e-02(8.21e-04)-	**1.5605e-01(3.99e-03)**
+/-/=			7/24/1	6/23/3	5/25/2	3/29/0	8/22/2	

Table A.4: The average HV values and standard deviations obtained by KE-MOEA and Five Competitive Algorithms on five real-world application problems

Problem	M	D	DKCA [42]	MOEA/PSL [34]	SparseEA [33]	S-ECSO [36]	S-NSGA-II [37]	KE-MOEA
SR1	2	1000	2.1941e-1 (6.17e-3) -	1.9188e-1 (3.38e-3) -	2.0832e-1 (3.39e-3) -	1.3253e-1 (0.00e+0) -	1.3253e-1 (1.00e-3) -	**2.5819e-1 (1.68e-3)**
SR2	2	2000	3.0729e-1 (8.19e-3) -	2.4941e-1 (3.83e-2) -	2.7618e-1 (9.10e-4) -	1.7023e-1 (0.00e+0) -	1.4023e-1 (2.00e-3) -	**3.8551e-1 (8.64e-3)**
SR3	2	4000	4.0192e-1 (9.05e-3) -	3.2790e-1 (1.80e-2) -	3.5497e-1 (2.03e-3) -	2.1202e-1 (0.00e+0) -	1.1863e-1 (7.00e-3) -	**4.7999e-1 (3.84e-3)**
KP1	2	1000	3.4836e-2 (4.09e-3) -	5.1168e-2 (4.73e-4) -	5.7643e-2 (2.12e-3) -	3.3370e-2 (0.00e+0) -	5.7043e-2 (2.12e-3) -	**5.8720e-2 (1.21e-4)**
KP2	2	2000	3.6126e-2 (1.95e-3) -	4.4985e-2 (3.22e-3) -	5.7460e-2 (8.82e-4) =	3.3536e-2 (0.00e+0) -	4.9985e-2 (3.22e-3) -	**5.7825e-2 (6.62e-4)**
KP3	2	4000	3.6633e-2 (2.61e-3) -	4.9225e-2 (9.96e-4) =	**5.7013e-2 (2.29e-4) +**	3.3503e-2 (0.00e+0) -	4.5718e-2 (7.55e-4) -	5.0800e-2 (1.40e-3)
NN1	2	435	8.4917e-1 (8.94e-3) =	8.5454e-1 (1.64e-2) =	8.5539e-1 (1.09e-2) =	8.1996e-1 (0.00e+0) -	**8.6320e-1 (1.21e-2) +**	8.5306e-1 (1.32e-2)
NN2	2	505	8.0534e-1 (7.28e-3) =	8.0867e-1 (1.09e-2) -	8.0537e-1 (1.08e-2) -	7.8483e-1 (0.00e+0) -	7.9302e-1 (7.78e-4) -	**8.2103e-1 (1.62e-2)**
NN3	2	937	8.0671e-1 (2.78e-3) -	**9.4827e-1 (1.13e-2) +**	9.2564e-1 (1.80e-2) +	7.3178e-1 (0.00e+0) -	8.8277e-1 (7.91e-3) =	8.7141e-1 (4.27e-2)
FS1	2	256	**9.2476e-1 (2.00e-3) +**	8.9319e-1 (9.84e-3) =	9.0036e-1 (1.83e-3) =	7.5250e-1 (1.92e-2) -	8.0996e-1 (7.23e-3) -	8.9056e-1 (8.21e-3)
FS2	2	310	9.8043e-1 (2.54e-2) -	9.8038e-1 (2.55e-2) -	9.7985e-1 (2.46e-2) -	9.6214e-1 (0.00e+0) -	9.6193e-1 (0.00e+0) -	**9.9797e-1 (1.45e-5)**
FS3	2	2308	**9.9971e-1 (0.00e+0) =**	9.2731e-1 (1.02e-1) -	9.9171e-1 (0.00e+0) =	9.9962e-1 (0.00e+0) =	9.9916e-1 (9.50e-5) =	9.9201e-1 (1.17e-3)
VR1	2	10201	6.0922e-01(4.69e-02) -	8.8317e-01(6.57e-02) -	3.7126e-01(3.58e-01) -	6.6528e-01(4.61e-02) -	1.2874e-01(1.18e-01) -	**9.8282e-1(2.43e-2)**
VR2	2	10201	6.4013e-01(1.78e-01) -	1.3096e-01(1.67e-01) -	5.6398e-01(1.18e-01) -	7.8334e-01(1.42e-02) -	2.7845e-01(1.49e-02) -	**9.8412e-1(1.59e-2)**
VR3	2	10201	3.4786e-01(1.98e-01) -	4.1414e-01(5.68e-01) -	3.6281e-01(5.48e-02) -	6.3267e-01(7.01e-02) -	1.7501e-01(1.15e-02) -	**9.8460e-1(3.36e-2)**
+/-/=			1/11/3	1/11/3	2/9/4	0/14/1	1/12/2	

Table A.5: The average HV values and standard deviations obtained by two SOTA algorithms and KE-MOEA on vehicle routing problems

Problem	M	D	MOALNS [44]	ACMODE [45]	KE-MOEA
VR1	2	10201	6.0284e-01(7.16e-02) -	5.0127e-01(3.68e-02) -	**9.8282e-1(2.43e-2)**
VR2	2	10201	7.8547e-01(1.85e-02) -	6.5087e-01(2.07e-02) -	**9.8412e-1(1.59e-2)**
VR3	2	10201	6.5753e-01(7.89e-02) -	3.4882e-01(7.12e-02) -	**9.8460e-1(3.36e-2)**
+/-/=			0/3/0	0/3/0	

Table A.6: IGD results obtained by four SOTA LSMOEAs and KE-MOEA on SMOP1-SMOP8 with 2 Objectives and 1000-10000 decision variables

Problem	M	D	TS-S-NSGA-II [37]	MOEA-NZD [32]	SparseEMT_S-NSGA-II [40]	KLEA [36]	KE-MOEA
SMOP1	2	1000	2.5253e-2 (7.76e-3) -	2.1258e-2 (3.24e-3) -	9.3235e-3 (5.20e-3) -	1.3835e-2 (1.63e-2) -	**3.7703e-3 (3.06e-5)**
SMOP1	2	2000	2.6629e-2 (4.64e-3) -	2.2552e-2 (3.01e-3) -	9.6755e-3 (5.10e-3) -	6.5427e-3 (1.03e-3) -	**5.7035e-3 (6.38e-4)**
SMOP1	2	4000	3.0969e-2 (3.91e-3) -	2.1433e-2 (6.79e-4) -	1.1629e-2 (6.05e-3) =	1.3492e-2 (6.79e-3) -	**1.0760e-2 (1.87e-3)**
SMOP1	2	10000	3.0636e-2 (5.19e-3) -	2.0336e-2 (3.66e-4) -	1.4013e-2 (9.93e-3) -	1.2630e-1 (1.55e-1) -	**4.7790e-3 (1.03e-4)**
SMOP2	2	1000	4.6393e-2 (9.24e-3) -	7.2183e-2 (1.82e-3) -	2.2259e-2 (1.22e-2) -	1.6777e-2 (4.83e-3) -	**1.3499e-2 (5.21e-3)**
SMOP2	2	2000	5.3505e-2 (7.00e-3) -	7.0224e-2 (4.41e-3) -	2.4219e-2 (3.98e-3) -	2.3691e-2 (2.93e-2) -	**2.1805e-2 (4.43e-3)**
SMOP2	2	4000	7.3608e-2 (1.05e-2) -	6.9886e-2 (5.82e-3) -	3.8231e-2 (1.63e-2) -	6.7764e-2 (6.98e-2) -	**3.3750e-2 (3.64e-3)**
SMOP2	2	10000	7.3383e-2 (7.46e-3) -	7.2701e-2 (2.23e-3) -	5.7079e-2 (1.69e-3) -	5.2572e-2 (4.25e-2) -	**1.9443e-2 (7.42e-4)**
SMOP3	2	1000	2.6081e-2 (7.83e-4) -	3.6740e-2 (5.27e-3) -	1.4053e-2 (4.89e-3) -	9.3349e-3 (8.79e-3) -	**3.7925e-3 (2.99e-5)**
SMOP3	2	2000	2.8984e-2 (1.01e-2) -	3.6730e-2 (2.51e-3) -	1.8346e-2 (1.26e-2) -	8.6773e-3 (1.08e-3) -	**8.1311e-3 (1.14e-3)**
SMOP3	2	4000	3.5112e-2 (6.47e-3) -	3.6018e-2 (1.77e-4) -	2.9863e-2 (5.28e-3) -	2.9142e-2 (3.24e-2) -	**2.4102e-2 (7.21e-3)**
SMOP3	2	10000	3.5581e-2 (5.12e-3) -	3.6327e-2 (5.13e-4) -	1.3684e-2 (1.64e-2) -	4.7824e-2 (3.61e-2) -	**8.6200e-3 (9.14e-4)**
SMOP4	2	1000	5.5267e-3 (1.00e-4) -	4.1496e-3 (1.25e-6) =	4.1747e-3 (1.66e-4) =	4.2781e-3 (9.11e-5) -	**4.1356e-3 (7.93e-5)**
SMOP4	2	2000	5.4405e-3 (1.82e-4) -	4.1500e-3 (1.65e-6) -	4.1990e-3 (9.36e-5) -	4.3042e-3 (4.22e-5) -	**4.1473e-3 (7.45e-5)**
SMOP4	2	4000	5.8936e-3 (3.96e-4) -	4.1501e-3 (2.47e-7) =	4.1975e-3 (8.53e-5) =	4.3388e-3 (8.35e-5) -	**4.1147e-3 (4.80e-5)**
SMOP4	2	10000	5.3534e-3 (1.28e-4) +	4.1528e-3 (3.49e-7) +	4.1784e-3 (6.62e-5) +	**4.1352e-3 (7.11e-5) +**	4.5031e-1 (1.12e-3)
SMOP5	2	1000	6.3341e-3 (8.11e-4) +	8.7719e-3 (4.01e-4) +	**4.2498e-3 (9.46e-5) +**	4.8160e-3 (9.07e-5) +	1.0471e-2 (6.25e-4)
SMOP5	2	2000	9.0662e-3 (1.71e-3) +	5.8587e-3 (4.27e-4) +	**4.3144e-3 (4.91e-5) +**	4.6526e-3 (8.15e-5) +	1.5595e-2 (1.50e-3)
SMOP5	2	4000	9.0215e-3 (3.98e-4) +	5.0200e-3 (3.19e-4) +	**4.3191e-3 (1.64e-4) +**	4.6512e-3 (1.83e-4) +	3.3847e-2 (3.22e-3)
SMOP5	2	10000	7.3176e-3 (2.02e-3) +	4.9663e-3 (9.82e-5) +	**4.1979e-3 (1.82e-4) +**	4.4308e-3 (1.19e-4) +	8.1213e-2 (6.85e-4)
SMOP6	2	1000	5.2884e-3 (2.57e-4) =	3.5920e-2 (1.12e-3) -	5.2344e-3 (4.14e-5) =	5.9989e-3 (1.87e-4) -	**5.1900e-3 (2.64e-4)**
SMOP6	2	2000	5.1093e-3 (3.03e-4) +	3.7239e-2 (2.14e-3) -	**4.9122e-3 (1.32e-4) +**	6.3146e-3 (2.18e-4) +	7.0231e-3 (2.53e-4)
SMOP6	2	4000	5.8108e-3 (3.41e-4) +	3.7779e-2 (1.17e-3) -	**4.8576e-3 (7.59e-5) +**	6.6996e-3 (1.58e-4) +	1.1141e-2 (2.89e-3)
SMOP6	2	10000	6.5750e-3 (3.13e-4) +	3.8130e-2 (3.43e-4) -	**4.9851e-3 (1.14e-4) +**	6.8003e-3 (2.54e-4) +	2.8535e-2 (6.64e-4)
SMOP7	2	1000	7.1496e-2 (1.24e-2) -	6.9427e-2 (4.79e-3) -	2.0217e-2 (1.31e-2) -	1.8038e-2 (1.73e-2) -	**1.7534e-2 (6.54e-3)**
SMOP7	2	2000	7.5599e-2 (2.18e-2) -	7.2094e-2 (5.32e-3) -	2.9695e-2 (2.85e-2) -	2.5766e-2 (1.47e-2) -	**2.4998e-2 (9.26e-3)**
SMOP7	2	4000	8.5754e-2 (1.48e-2) -	8.0680e-2 (4.28e-3) -	5.2204e-2 (1.68e-2) -	6.7635e-2 (9.30e-2) -	**3.9887e-2 (1.84e-2)**
SMOP7	2	10000	8.0089e-2 (2.85e-3) -	8.3519e-2 (5.73e-3) -	5.4192e-2 (3.48e-2) -	4.3903e-2 (6.21e-2) -	**2.6436e-2 (8.68e-4)**
SMOP8	2	1000	2.4959e-1 (3.30e-2) -	2.5997e-1 (1.46e-2) -	2.7392e-1 (4.78e-3) -	1.6679e-1 (1.07e-2) -	**1.1062e-1 (8.10e-3)**
SMOP8	2	2000	2.4864e-1 (9.66e-3) -	2.6732e-1 (1.02e-2) -	2.9297e-1 (1.51e-2) -	1.9145e-1 (3.23e-2) -	**1.2590e-1 (6.10e-3)**
SMOP8	2	4000	2.9140e-1 (9.34e-3) -	2.7135e-1 (1.98e-3) -	3.0142e-1 (2.36e-2) -	2.5248e-1 (1.36e-1) -	**1.4989e-1 (6.83e-3)**
SMOP8	2	10000	3.0204e-1 (4.12e-3) -	2.7051e-1 (4.50e-3) -	3.0159e-1 (2.54e-2) -	1.6676e-1 (5.59e-3) -	**1.5051e-1 (4.39e-3)**
+/-/=			8/23/1	5/25/2	8/20/4	8/24/0	

Table A.7: HV results obtained by four SOTA LSMOEAs and KE-MOEA on SMOP1-SMOP8 with 2 objectives and 1000-10000 decision variables

Problem	M	D	TS-S-NSGA-II [37]	MOEA-NZD [32]	SparseEMT_S-NSGA-II [40]	KLEA [36]	KE-MOEA
SMOP1	2	1000	5.5193e-1 (9.73e-3) -	5.5759e-1 (4.04e-3) -	5.7337e-1 (7.99e-3) -	5.6790e-1 (2.17e-2) -	**5.8186e-1 (8.07e-5)**
SMOP1	2	2000	5.5027e-1 (5.82e-3) -	5.5602e-1 (3.73e-3) -	5.7285e-1 (7.78e-3) -	5.7661e-1 (1.61e-3) -	**5.7779e-1 (1.04e-3)**
SMOP1	2	4000	5.4481e-1 (4.86e-3) -	5.5743e-1 (8.42e-4) -	5.6359e-1 (8.64e-3) -	5.6158e-1 (9.24e-3) -	**5.6947e-1 (2.16e-3)**
SMOP1	2	10000	5.4426e-1 (6.30e-3) -	5.5879e-1 (4.58e-4) -	5.6678e-1 (1.27e-2) -	4.0565e-1 (2.30e-1) -	**5.7934e-1 (2.07e-4)**
SMOP2	2	1000	5.2528e-1 (1.18e-2) -	4.9194e-1 (2.56e-3) -	5.5638e-1 (1.54e-2) -	5.6099e-1 (7.00e-3) -	**5.6731e-1 (6.63e-3)**
SMOP2	2	2000	5.1612e-1 (9.09e-3) -	4.9537e-1 (5.69e-3) -	5.5018e-1 (5.13e-3) -	5.5372e-1 (3.82e-2) -	**5.5686e-1 (5.48e-3)**
SMOP2	2	4000	4.9050e-1 (1.35e-2) -	4.9616e-1 (7.49e-3) -	5.3630e-1 (2.04e-2) -	4.9970e-1 (8.73e-2) -	**5.3899e-1 (2.64e-3)**
SMOP2	2	10000	4.8982e-1 (9.10e-3) -	4.9257e-1 (2.87e-3) -	5.1246e-1 (2.16e-3) -	5.1780e-1 (5.43e-2) -	**5.5981e-1 (9.26e-4)**
SMOP3	2	1000	5.5089e-1 (8.22e-4) -	5.3812e-1 (6.68e-3) -	5.6654e-1 (6.23e-3) -	5.7359e-1 (1.21e-2) -	**5.8178e-1 (1.32e-4)**
SMOP3	2	2000	5.4732e-1 (1.27e-2) -	5.3835e-1 (3.15e-3) -	5.6210e-1 (1.73e-2) -	5.7088e-1 (2.13e-3) -	**5.7424e-1 (1.58e-3)**
SMOP3	2	4000	5.3940e-1 (8.17e-3) -	5.3931e-1 (2.23e-4) -	5.4483e-1 (7.91e-3) -	5.4509e-1 (4.13e-2) -	**5.4884e-1 (5.61e-3)**
SMOP3	2	10000	5.3767e-1 (6.59e-3) -	5.3894e-1 (6.43e-4) -	5.6827e-1 (2.17e-2) -	5.2409e-1 (4.60e-2) -	**5.7357e-1 (1.22e-3)**
SMOP4	2	1000	8.1741e-1 (7.51e-5) -	8.1785e-1 (4.40e-6) -	8.1872e-1 (1.37e-4) -	8.1865e-1 (6.56e-5) -	**8.1921e-1 (4.97e-5)**
SMOP4	2	2000	8.1749e-1 (1.71e-4) -	8.1786e-1 (7.20e-6) -	8.1875e-1 (4.42e-5) -	8.1870e-1 (2.31e-5) -	**8.1920e-1 (4.37e-5)**
SMOP4	2	4000	8.1720e-1 (2.74e-4) -	8.1788e-1 (3.48e-6) -	8.1875e-1 (1.12e-4) =	8.1867e-1 (4.69e-5) -	**8.1917e-1 (2.34e-5)**
SMOP4	2	10000	8.1762e-1 (6.10e-5) +	8.1791e-1 (2.92e-7) +	**8.1876e-1 (5.86e-5) +**	8.1868e-1 (4.89e-5) +	3.1040e-1 (2.42e-3)
SMOP5	2	1000	8.1453e-1 (1.24e-3) +	8.1016e-1 (3.96e-4) =	**8.1750e-1 (1.12e-4) +**	8.1691e-1 (2.77e-4) +	8.0959e-1 (7.34e-4)
SMOP5	2	2000	8.1083e-1 (2.15e-3) +	8.1416e-1 (2.50e-4) +	**8.1747e-1 (1.35e-4) +**	8.1695e-1 (1.37e-4) +	8.0380e-1 (1.67e-3)
SMOP5	2	4000	8.1079e-1 (4.89e-4) +	8.1538e-1 (4.42e-4) +	**8.1762e-1 (1.62e-4) +**	8.1704e-1 (2.62e-4) +	7.8152e-1 (2.43e-3)
SMOP5	2	10000	8.1308e-1 (2.70e-3) +	8.1545e-1 (1.39e-4) +	**8.1775e-1 (3.74e-4) +**	8.1740e-1 (1.40e-4) +	7.3049e-1 (8.01e-4)
SMOP6	2	1000	**8.1694e-1 (2.69e-4) =**	7.7938e-1 (1.05e-3) -	8.1635e-1 (3.34e-5) =	8.1476e-1 (2.64e-4) -	8.1644e-1 (4.72e-4)
SMOP6	2	2000	**8.1646e-1 (4.28e-4) +**	7.7823e-1 (2.19e-3) -	8.1633e-1 (2.08e-4) +	8.1427e-1 (3.24e-4) +	8.1379e-1 (3.15e-4)
SMOP6	2	4000	8.1538e-1 (5.91e-4) +	7.7772e-1 (1.25e-3) -	**8.1633e-1 (1.43e-4) +**	8.1369e-1 (2.17e-4) +	8.0648e-1 (3.22e-4)
SMOP6	2	10000	8.1481e-1 (9.02e-4) +	7.7741e-1 (3.44e-4) -	**8.1617e-1 (1.70e-4) +**	8.1358e-1 (2.89e-4) +	7.8959e-1 (7.43e-4)
SMOP7	2	1000	2.5060e-1 (1.74e-2) -	2.5341e-1 (7.35e-3) -	3.2206e-1 (1.79e-2) -	3.2513e-1 (2.47e-2) -	**3.2561e-1 (8.85e-3)**
SMOP7	2	2000	2.4476e-1 (3.06e-2) -	2.4971e-1 (7.64e-3) -	3.0893e-1 (3.92e-2) -	3.1449e-1 (2.03e-2) -	**3.1526e-1 (1.27e-2)**
SMOP7	2	4000	2.3054e-1 (2.10e-2) -	2.3773e-1 (6.08e-3) -	2.7799e-1 (2.31e-2) -	2.6540e-1 (1.14e-1) -	**3.0300e-1 (3.16e-2)**
SMOP7	2	10000	2.3795e-1 (4.14e-3) -	2.3376e-1 (8.09e-3) -	2.7509e-1 (4.79e-2) -	2.8994e-1 (8.57e-2) -	**3.1319e-1 (1.15e-3)**
SMOP8	2	1000	8.3164e-2 (1.87e-2) -	7.4595e-2 (8.27e-3) -	6.8723e-2 (2.31e-3) -	1.4055e-1 (9.72e-3) -	**1.9763e-1 (9.73e-3)**
SMOP8	2	2000	8.2834e-2 (5.43e-3) -	7.1085e-2 (5.64e-3) -	5.9261e-2 (7.60e-3) -	1.2242e-1 (2.57e-2) -	**1.8042e-1 (6.46e-3)**
SMOP8	2	4000	5.9853e-2 (4.34e-3) -	6.9382e-2 (1.15e-3) -	5.5625e-2 (1.12e-2) -	9.5797e-2 (6.92e-2) -	**1.5431e-1 (4.41e-3)**
SMOP8	2	10000	5.3935e-2 (1.75e-3) -	7.0043e-2 (2.44e-3) -	5.5818e-2 (1.15e-2) -	1.4131e-1 (5.00e-3) -	**1.5605e-1 (3.99e-3)**
+/-/=			8/23/1	4/27/1	8/22/2	8/22/2	

Table A.8: The average HV values and standard deviations obtained by four SOTA LSMOEAs on five real-world application problems

Problem	M	D	TS-S-NSGA-II [37]	MOEA-NZD [32]	SparseEMT_S-NSGA-II [40]	KLEA [36]	KE-MOEA
SR1	2	1000	1.0436e-1 (9.49e-4) -	2.4509e-1 (1.34e-4) -	1.1167e-1 (1.04e-3) -	2.4347e-1 (2.46e-3) -	**2.5819e-1 (1.68e-3)**
SR2	2	1000	1.0436e-1 (9.49e-4) -	2.7509e-1 (1.34e-4) -	1.1167e-1 (1.04e-3) -	3.7347e-1 (2.46e-3) -	**3.8551e-1 (8.64e-3)**
SR3	2	1000	1.0436e-1 (9.49e-4) -	2.7509e-1 (1.34e-4) -	1.1167e-1 (1.04e-3) -	3.7347e-1 (2.46e-3) -	**4.7999e-1 (3.84e-3)**
KP1	2	1000	5.7414e-2 (1.98e-3) -	5.7201e-2 (8.59e-5) -	4.4567e-2 (2.78e-3) -	5.8334e-2 (2.34e-4) =	**5.8720e-2 (1.21e-4)**
KP2	2	2000	5.5202e-2 (6.25e-4) -	5.6319e-2 (4.47e-4) -	4.3265e-2 (1.71e-3) -	5.2279e-2 (1.89e-3) -	**5.7825e-2 (6.62e-4)**
KP3	2	4000	4.9964e-2 (1.08e-3) -	**5.6358e-2 (4.08e-4) +**	3.7062e-2 (6.24e-4) -	5.1637e-2 (4.45e-4) =	5.0800e-2 (1.40e-3)
NN1	2	435	8.6300e-1 (1.07e-2) +	8.5547e-1 (1.39e-2) =	8.3057e-1 (2.64e-2) -	**8.6371e-1 (1.29e-2) +**	8.5306e-1 (1.32e-2)
NN2	2	505	7.9912e-1 (1.11e-2) -	8.1568e-1 (1.66e-4) -	8.1811e-1 (8.60e-3) -	8.1014e-1 (3.26e-3) -	**8.2103e-1 (1.62e-2)**
NN3	2	937	8.0123e-1 (1.87e-6) -	8.0433e-1 (6.34e-4) -	8.0659e-1 (3.75e-3) -	8.1149e-1 (1.50e-3) -	**8.7141e-1 (4.27e-2)**
FS1	2	256	8.0608e-1 (3.26e-2) -	8.3855e-1 (2.30e-2) -	9.2449e-1 (5.76e-3) +	**9.3327e-1 (1.81e-3) +**	8.9056e-1 (8.21e-3)
FS2	2	310	9.9613e-1 (3.77e-4) =	9.9656e-1 (7.54e-5) =	9.8845e-1 (7.54e-5) -	9.8819e-1 (3.02e-4) -	**9.9797e-1 (1.45e-5)**
FS3	2	2308	9.9960e-1 (6.33e-5) =	**9.9971e-1 (0.00e+0) =**	9.9951e-1 (1.27e-4) =	9.9970e-1 (1.58e-5) =	9.9201e-1 (1.17e-3)
VR1	2	10201	5.4634e-02(5.36e-02) -	9.4117e-01(2.19e-02) -	3.2724e-02(1.95e-02) -	4.4549e-01(6.76e-02) -	**9.8282e-1(2.43e-2)**
VR2	2	10201	1.8189e-01(8.33e-02) -	9.3422e-01(1.98e-02) -	3.6796e-02(2.94e-02) -	6.1894e-01(1.19e-01) -	**9.8412e-1(1.59e-2)**
VR3	2	10201	1.3978e-01(1.72e-01) -	9.4790e-01(5.68e-03) -	6.5649e-02(6.38e-02) -	5.6660e-01(2.11e-02) -	**9.8460e-1(3.36e-2)**
+/-/=			1/12/2	1/11/3	1/13/1	2/10/3	

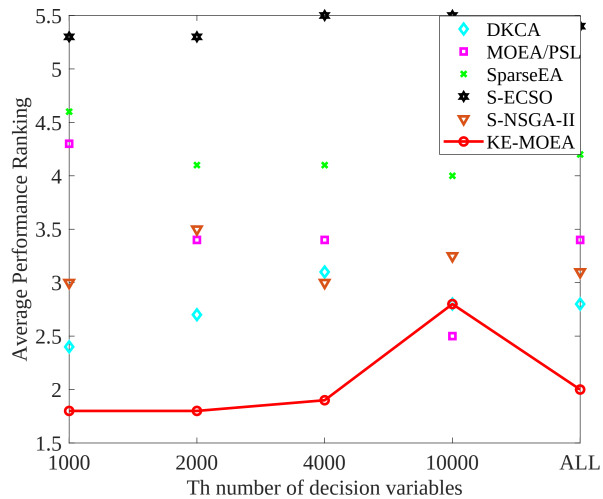

Fig. A.1: The performance ranking of all algorithms in solving all test problems for different dimensions of variables.

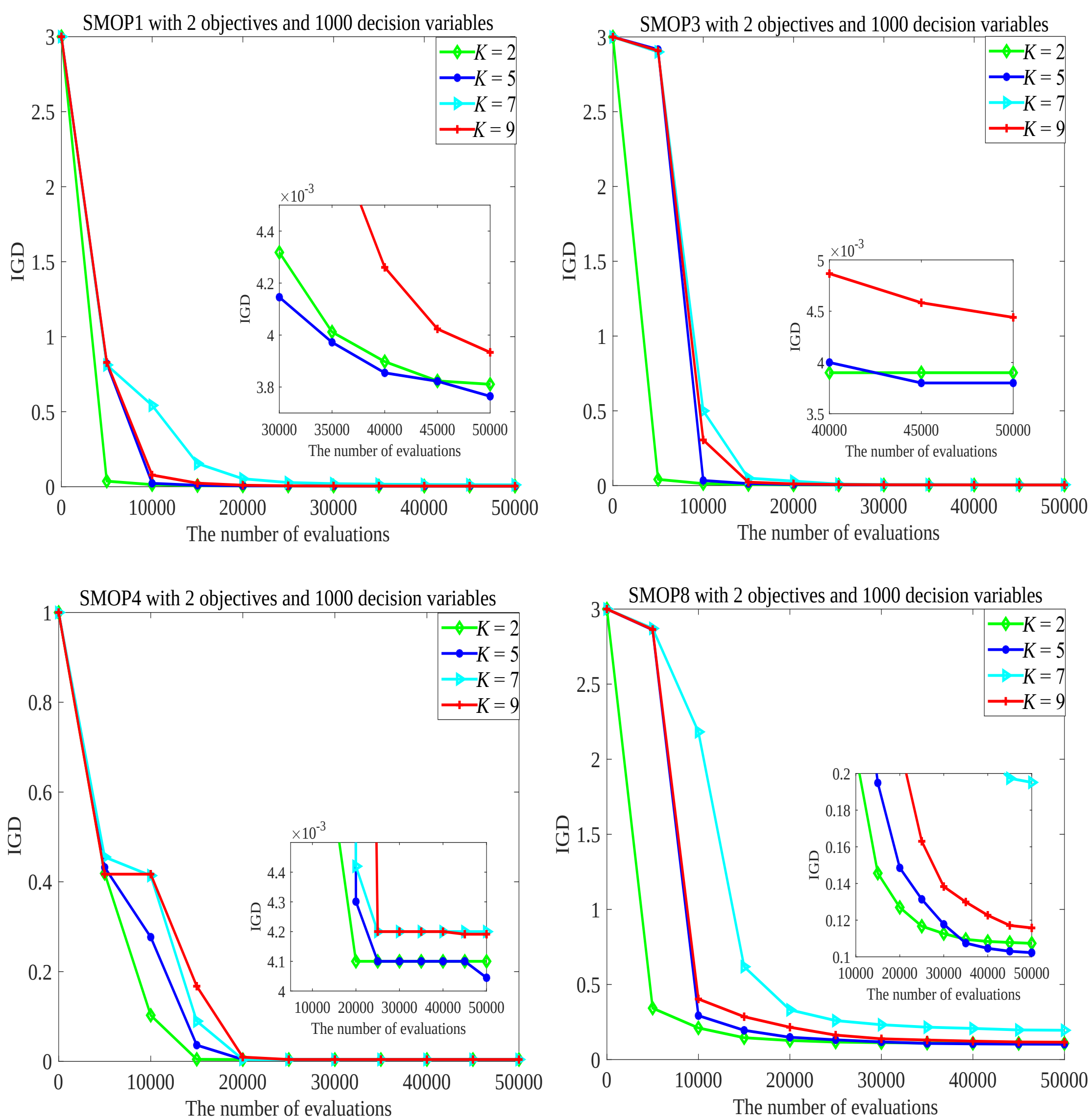

Fig. A.2: Parameter sensitivity analysis of K.

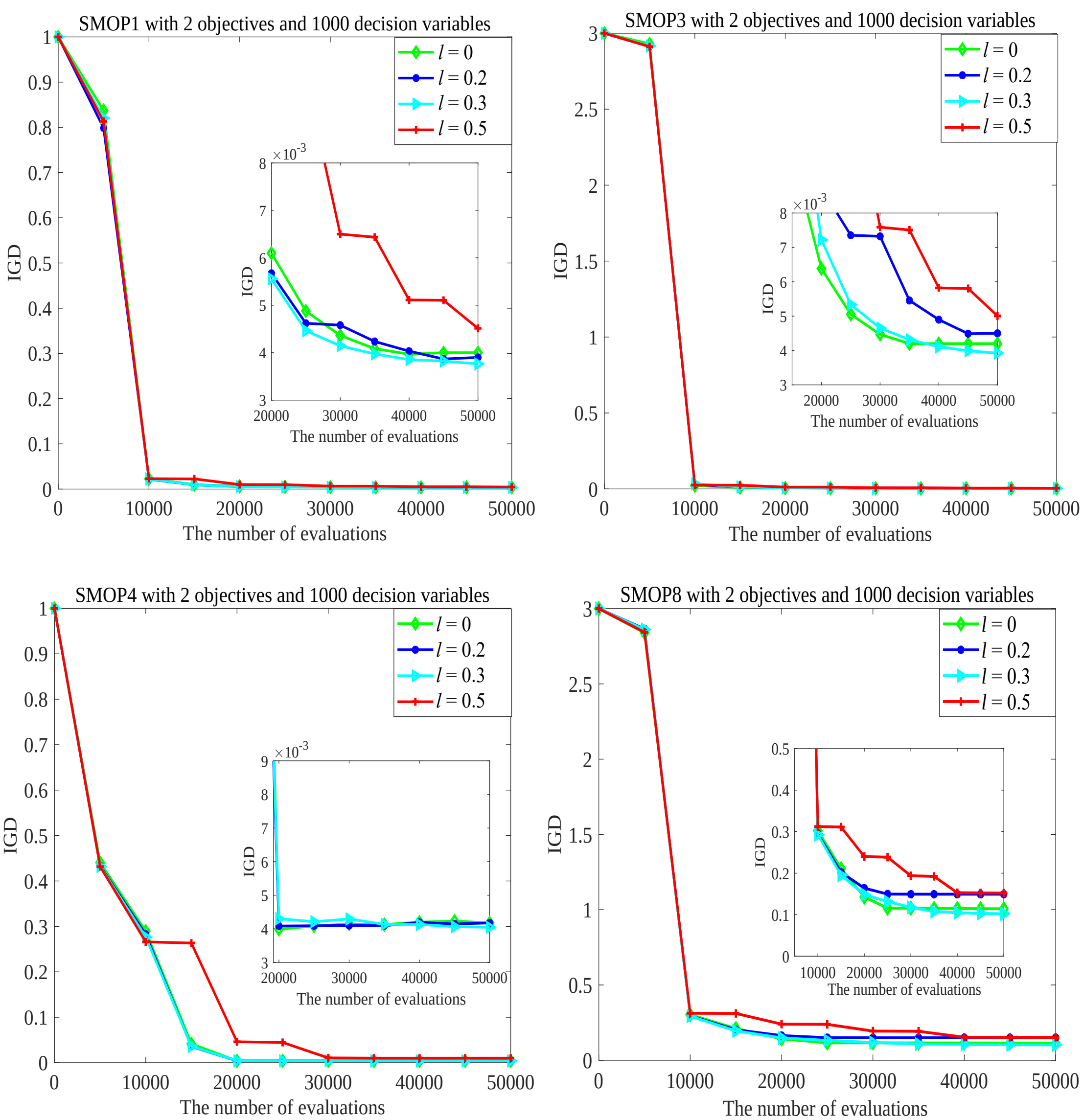

Fig. A.3: Parameter sensitivity analysis of l.

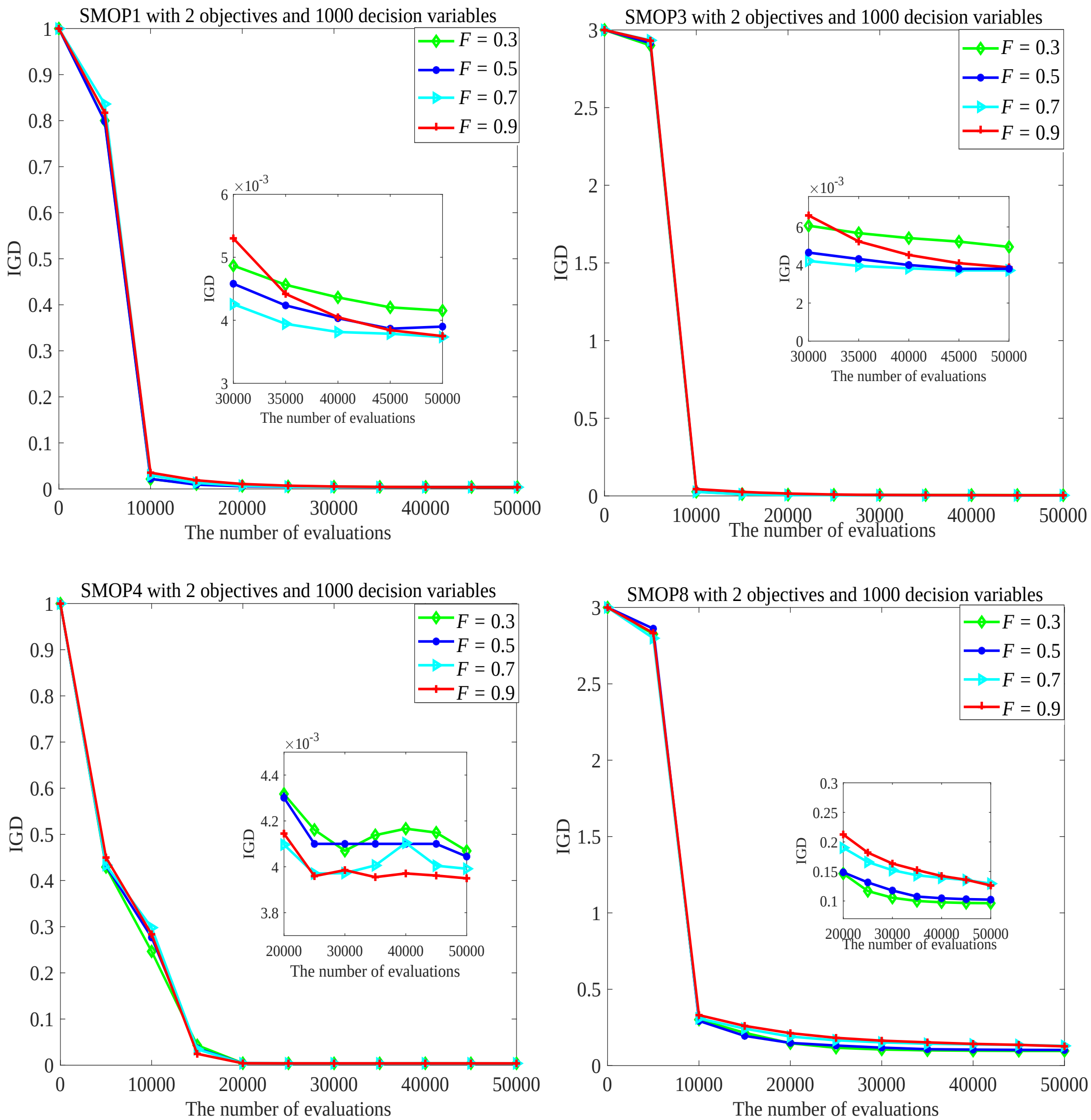

Fig. A.4: Parameter sensitivity analysis of F.

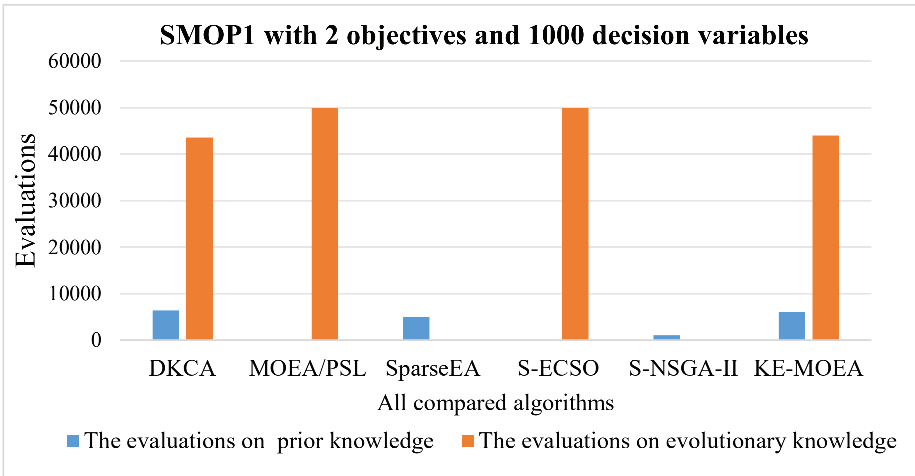

Fig. A.5: The consumed evaluation resources of all algorithms in exploring various knowledge when solving SMOP1 with 2 objectives with 1000 decision variables.

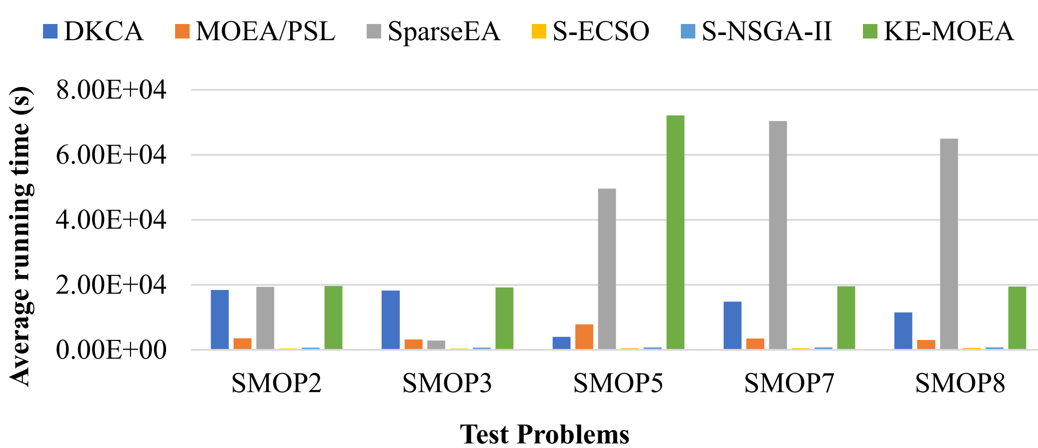

Fig. A.6: The CPU running time of all compared algorithms in solving SMOP test problems.

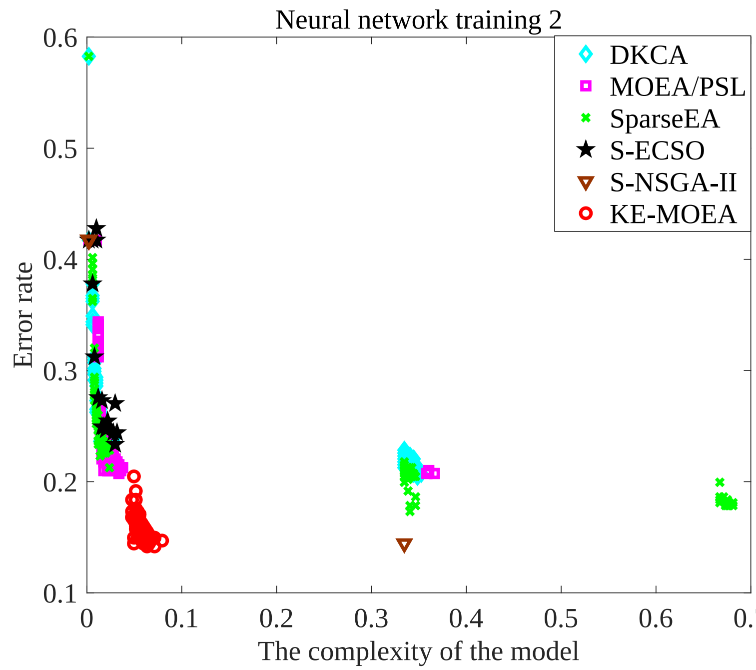

Fig. A.7: The final solutions obtained by all compared algorithms on NN2.

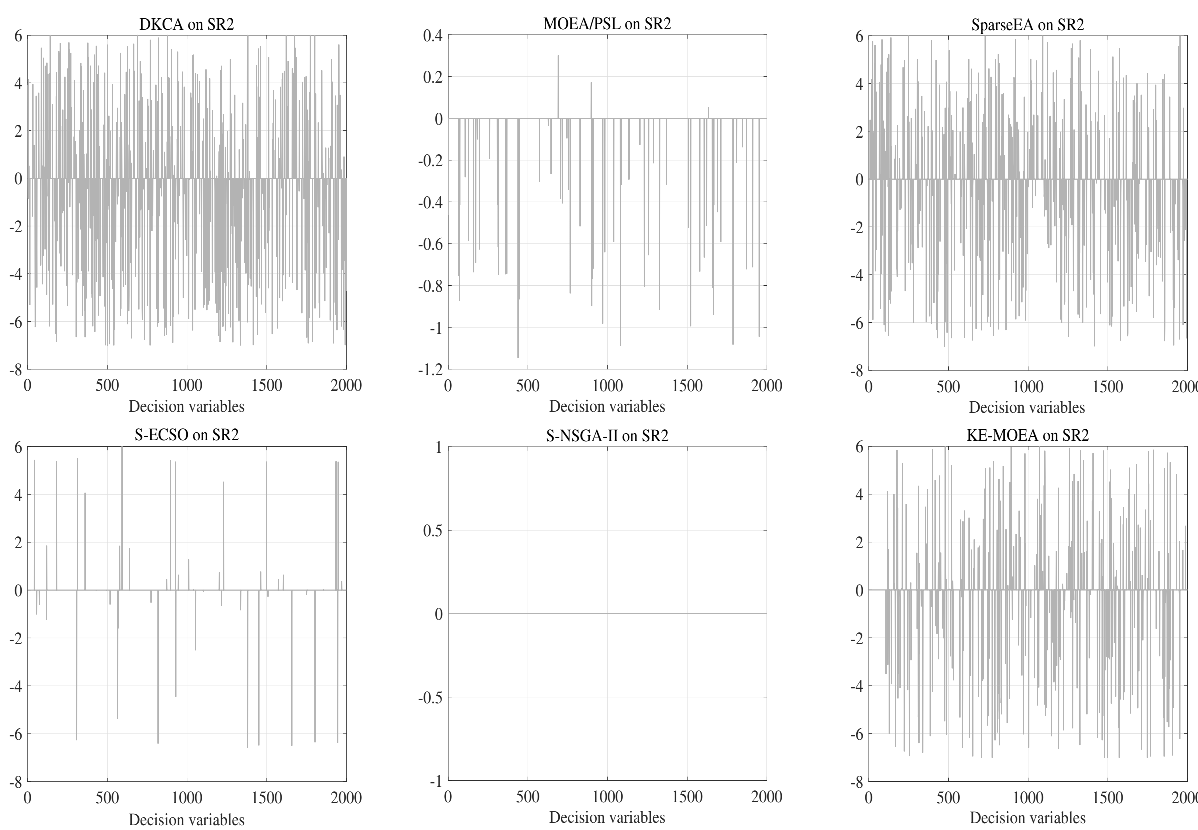

Fig. A.8: The sparsity patterns of decision variables obtained by all compared algorithms in solving SR2 problem.

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